

SODERARM, Sweden

WMO: 024930

Lat: 59.753N

Lon: 19.405E

Elev: 46

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.9	18.8	9.2	8.8	17.4	12.2	10.3	20.2	43.6	30.7	40.9	28.8	18.9	60	0.860

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	72.7	67.5	70.3	65.7	68.1	63.7	68.9	71.7	66.4	69.4	64.4	67.3	11.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.2	71.0	65.2	93.2	68.5	63.0	86.2	66.3	33.2	72.2	31.1	69.5	29.5	67.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
40.3	35.7	32.0	DB	13.8	74.8	7.5	2.8	8.4	76.8	3.9	78.5	-0.3	80.1	-5.8	82.1
			WB	13.2	68.6	4.6	2.4	9.9	70.3	7.2	71.7	4.7	73.0	1.3	74.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	31.1	30.1	32.8	38.4	46.2	55.1	62.7	63.0	55.8	47.0	40.1	35.1
	DBStd	12.47	5.81	6.17	5.05	3.66	4.08	4.34	4.32	3.79	3.84	4.74	4.80	5.84
	HDD50	3045	587	558	534	348	130	8	0	0	5	115	297	463
	HDD65	7400	1052	978	999	798	583	297	99	84	277	558	747	928
	CDD50	1169	0	0	0	0	12	160	394	403	178	22	0	0
	CDD65	50	0	0	0	0	0	0	28	22	0	0	0	0
	CDH74	28	0	0	0	0	0	0	21	7	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.9	20.4	18.4	17.1	15.5	14.6	14.2	13.6	14.0	16.3	18.5	19.9	20.8
Precipitation	PrecAvg	19.5	1.5	1.1	0.9	1.1	1.3	1.9	1.8	2.2	1.9	1.9	2.0	1.8
	PrecMax	28.4	3.1	2.1	2.0	2.6	3.0	3.7	4.6	5.6	4.6	4.3	4.2	3.7
	PrecMin	14.4	0.1	0.2	0.0	0.2	0.3	0.6	0.2	0.0	0.2	0.0	0.6	0.4
	PrecStd	3.0	0.7	0.5	0.5	0.7	0.7	0.8	1.1	1.3	1.1	0.9	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	42.0	45.0	51.0	59.3	67.4	76.5	74.7	65.7	58.2	50.3	47.6
		MCWB	40.6	38.5	40.6	44.7	52.0	60.0	68.7	69.9	61.4	56.1	48.4	44.9
	2%	DB	41.1	39.9	42.3	47.4	56.0	65.1	73.6	72.2	63.7	55.9	48.7	44.9
		MCWB	39.4	37.4	38.8	42.9	50.7	59.1	68.0	67.0	59.8	53.5	47.2	43.0
	5%	DB	39.8	38.5	40.3	45.3	53.7	63.0	71.4	70.1	62.1	54.5	47.4	43.4
		MCWB	38.3	36.5	37.6	41.7	49.4	58.3	66.9	65.5	58.7	52.3	45.9	41.8
	10%	DB	38.3	37.0	38.6	43.3	51.7	61.3	69.2	68.1	60.9	53.2	46.1	42.0
		MCWB	36.8	35.4	36.4	40.4	48.4	57.3	65.0	63.7	57.8	51.3	44.7	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	39.6	41.3	45.5	53.8	62.3	71.8	71.9	62.6	56.8	49.3	44.9
		MCDB	42.5	40.8	44.0	49.6	57.0	65.2	73.4	73.8	64.2	57.5	49.7	45.7
	2%	WB	39.9	38.0	39.3	43.4	51.6	60.6	69.6	68.8	60.8	54.1	47.6	43.0
		MCDB	40.8	39.3	41.7	46.5	54.7	63.2	72.6	71.2	62.8	55.1	48.4	43.7
	5%	WB	38.5	36.8	37.9	42.1	50.3	59.3	67.4	66.3	59.4	52.9	46.3	41.5
		MCDB	39.6	38.2	39.7	44.6	53.0	62.0	70.5	69.1	61.5	54.1	47.2	42.5
	10%	WB	37.0	35.7	36.8	40.8	48.9	58.0	65.2	64.6	58.4	51.7	45.0	40.3
		MCDB	38.3	37.0	38.3	43.0	51.3	60.8	68.3	67.5	60.5	53.0	45.9	41.5
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.0	5.2	5.9	6.2	6.4	5.9	5.3	4.8	4.4	4.3	5.0
		MCDBR	5.2	5.2	7.1	9.1	9.3	9.1	8.0	6.5	5.7	5.0	4.5	5.1
		MCWBR	5.0	4.8	5.1	5.6	5.8	5.3	4.7	4.5	4.2	4.6	4.6	4.7
	5% WB	MCDBR	5.1	5.0	6.5	8.6	8.4	8.0	7.1	5.8	5.0	4.9	3.9	4.9
		MCWBR	5.1	4.9	5.0	5.6	5.8	5.1	4.9	4.6	4.1	4.9	4.2	4.9
Clear-Sky Solar Irradiance	taub	0.282	0.317	0.335	0.373	0.369	0.361	0.389	0.379	0.348	0.335	0.307	0.265	
	taud	2.210	2.284	2.340	2.335	2.421	2.487	2.427	2.443	2.513	2.416	2.237	2.120	
	Ebn at Noon	159	217	253	262	273	277	265	258	249	210	145	115	
	Edh at Noon	13	22	28	33	33	31	32	30	23	19	13	10	
All-Sky Solar Radiation	RadAvg	83	254	732	1271	1713	1905	1778	1394	879	377	113	56	
	RadStd	10	48	75	115	138	112	165	115	91	60	26	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-198	-236	N/A	N/A

Nomenclature: See separate page